## STAT3612 Group Project Proposal: Multimodal Presurgical Brain Tumor Classification

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### 1. Project Overview

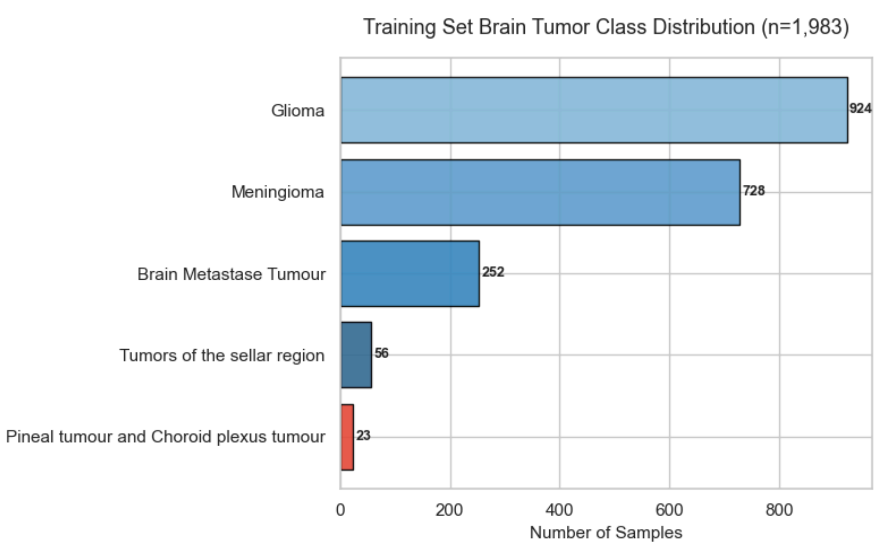
This project develops statistical machine learning models for presurgical brain tumor classification into five categories (Glioma, Meningioma, Brain Metastasis, Sellar-region tumours, Pineal/Choroid plexus tumours) using multimodal clinical data. Our approach follows a **three-stage pipeline**: (1) single-modality baselines to understand each modality's contribution, (2) multimodal fusion comparing early and late strategies, and (3) model selection with rigorous evaluation, class imbalance mitigation, and interpretability analysis.

We leverage all four available modalities — deep image features (ResNet-50, 2048-d per MRI sequence), radiomics (5 features × 4 sequences), structured clinical information (demographics, signal intensities, tumor location), and radiology report text. The primary evaluation metric is F1 metric, consistent with the Kaggle competition. The dataset contains 1,983 training / 283 validation / 572 test cases, with severe class imbalance (~40:1 between the largest and smallest classes).

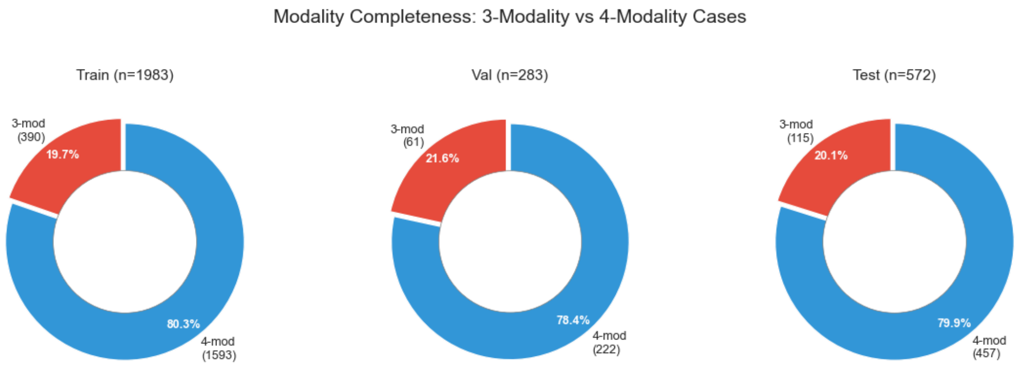
### 2. Preliminary Findings

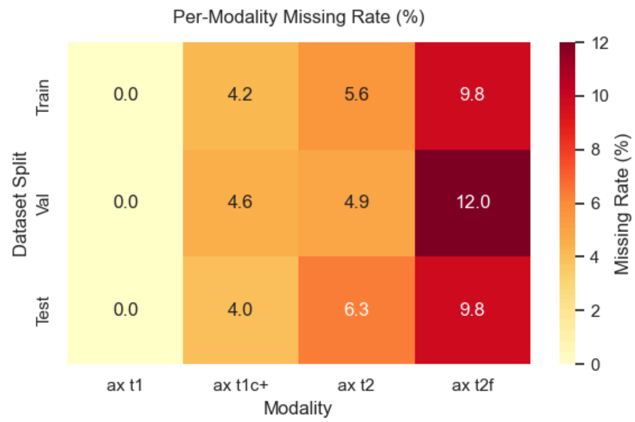
Initial EDA and a baseline experiment have already been completed:

- **Class imbalance**: Glioma (924 cases, 46.6%) and Meningioma (728, 36.7%) dominate, while Pineal/Choroid (23, 1.2%) and Sellar-region (56, 2.8%) are severely underrepresented. We adopt class_weight='balanced' as the default mitigation.

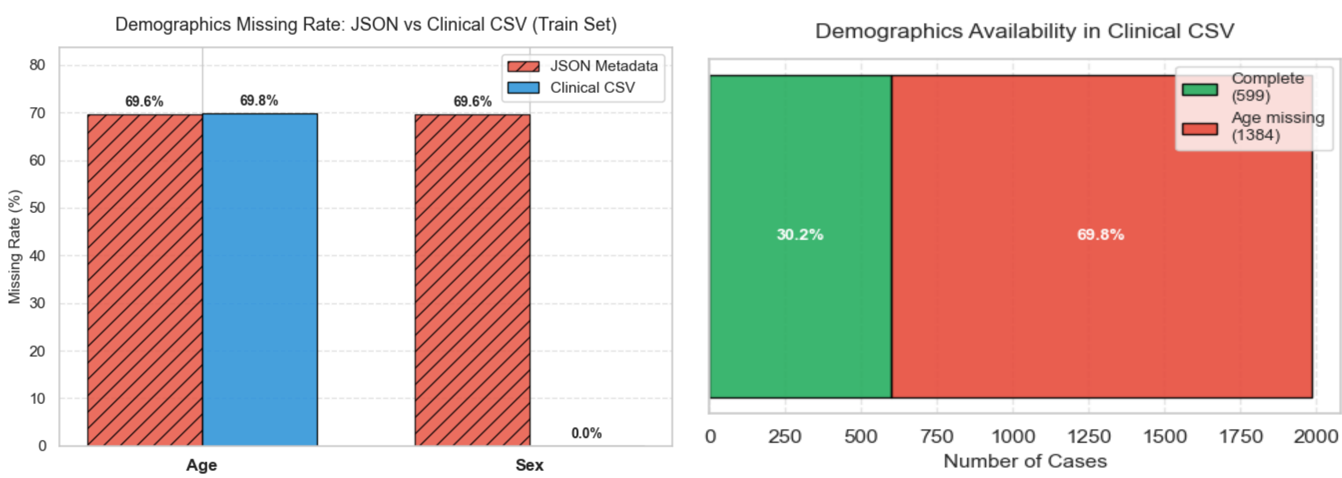



- **Multimodal incompleteness**: ~20% of cases lack one MRI modality; we zero-pad missing feature vectors as structural missingness.



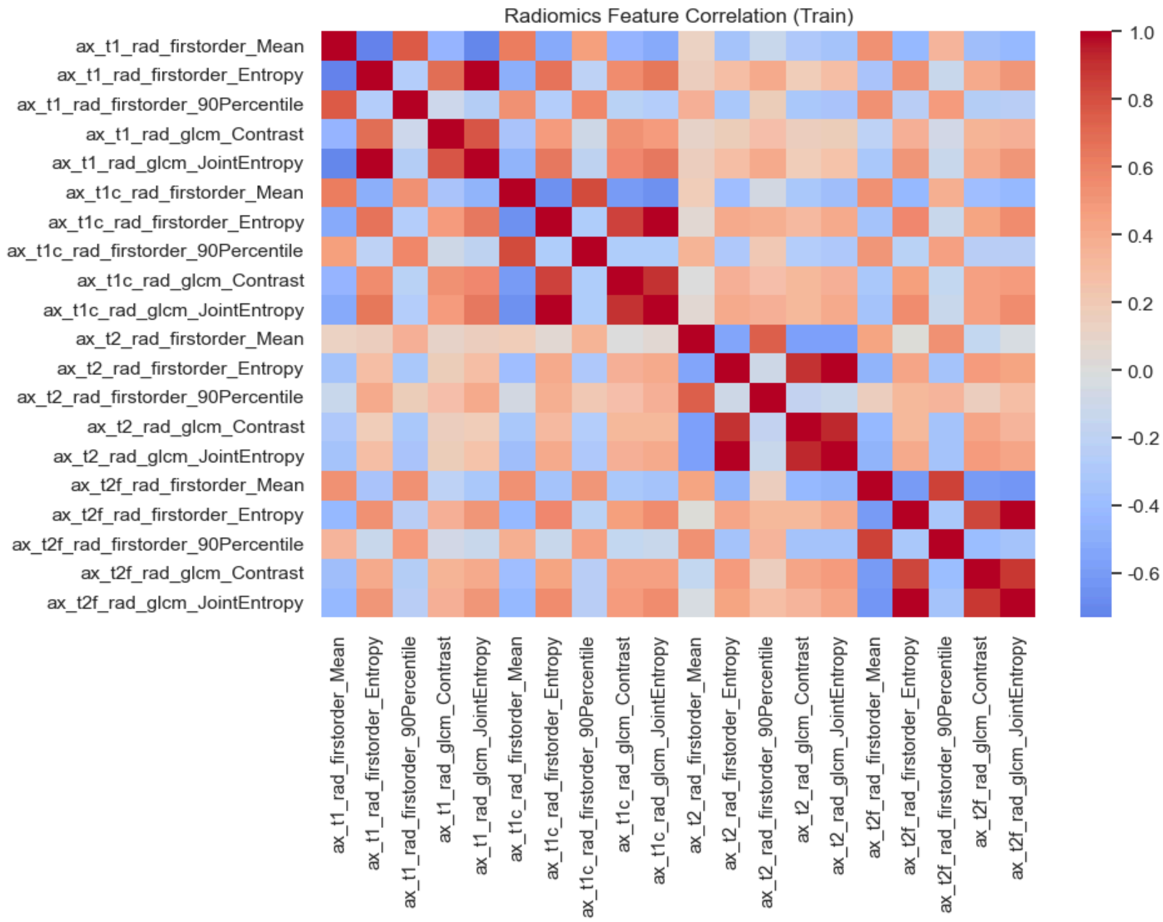




· **Missing demographics**: JSON metadata has ~70% missing for Age/Sex; we use clinical_information CSVs (fully populated) as the single source of truth.




· **Baseline result**: A Multinomial Logistic Regression (L2, class_weight='balanced') on the fused 800-d feature set achieves a validation macro-F1 of $0.6946 \pm 0.0523$. A preliminary modality ablation using LightGBM confirmed that fusing all four modalities outperforms any single modality.

· **Key insight**: Class weighting alone does not sufficiently lift minority-class F1; the high dimensionality (800-d vs 1,983 samples) creates overfitting risk requiring regularization and feature selection.

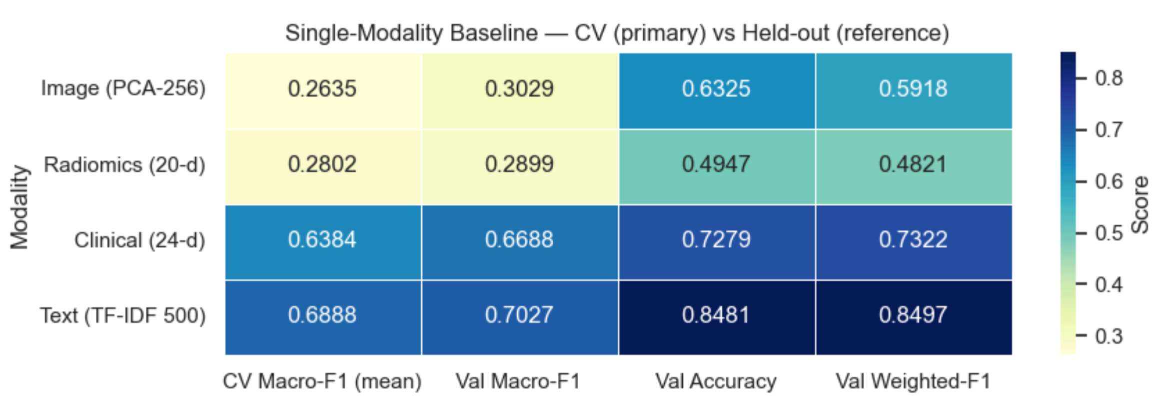



## 3. Proposed Methods

### Stage 1: Single-Modality Baselines

Before fusion, we build independent classifiers for each modality to understand their individual discriminative power:

- **Deep image features (8192-d → PCA-256)**: Logistic Regression and XGBoost to test linear vs non-linear separability.
- **Radiomics (20-d)**: feature selection via Mutual Information ranking (removing |r| > 0.95 pairs), followed by RF and SVM.
- **Clinical features (24-d)**: encoded demographics, signal intensities, and location keywords; LDA/QDA (Lecture 3.2) to test generative vs discriminative classifiers.
- **Text features (TF-IDF 500-d)**: Logistic Regression with L1 penalty (Lecture 4.2) for simultaneous regularization and feature selection.



### Stage 2: Multimodal Fusion

We systematically compare two canonical fusion strategies:

- **Early fusion**: After standardization and PCA (applied to image features), all modality-specific features are concatenated into a unified 800-dimensional vector (256 image + 20 radiomics + 24 clinical + 500 text). Classifiers (LR, RF, XGBoost, SVM-RBF, LightGBM, and MLPs) are trained on this combined space to assess whether a single model can effectively capture cross-modality interactions.

- **Late fusion**: Optimal single-modality models from Stage 1 are trained independently. Their predictive probabilities are then aggregated via (a) weighted soft voting and (b) a stacking meta-learner (using Logistic Regression as the second-level model). This approach preserves modality-specific decision boundaries while leveraging ensemble diversity.

Most models are selected from a principled set grounded in course theory: - LR with L1/L2 (Lecture 4.2): linear baseline with regularization and implicit feature selection. - LDA/QDA (Lecture 3.2): generative classifiers testing linear vs quadratic decision boundaries. - Random Forest (Lecture 5.1): bagging-based ensemble reducing variance via bootstrap aggregation. - XGBoost/LightGBM (Lecture 5.2): boosting-based ensemble reducing bias by sequentially focusing on misclassified examples. - SVM-RBF (Lecture 7): kernel trick for non-linear separation in high-dimensional feature space. In addition, we will evaluate the performance of deep neural networks, such as Multilayer Perceptrons (MLPs), on our datasets to explore the potential of representation learning.

Model and hyperparameter selection will rely on **stratified 5-fold cross-validation** (Lecture 4.1), selecting the best *procedure* and retraining on the full training set for validation.

### Stage 3: Optimization and Class Imbalance

- **Feature selection**: compare LASSO-based sparsity (L1), tree-based importance (RF/XGBoost), and MI ranking for identifying informative features. Optimal feature subsets are determined via nested cross-validation using validation macro-F1.

- **PCA tuning**: evaluate the trade-off between cumulative explained variance and downstream classification performance (macro-F1) for the deep image feature block.

- **Class imbalance**: conduct a controlled comparison of (a) class_weight only, (b) random oversampling, and (c) SMOTE applied in the PCA-reduced feature space. Performance is evaluated on the held-out validation set using per-class F1 and macro-F1 with particular focus on the two rarest classes: Pineal/Choroid plexus tumour and Tumors of the sellar region.

## 4. Model Analysis Plan

Focusing on the model analysis of the project, we design the following analysis content to target the workload:

- **Model comparison**: Evaluate all candidate models on the validation set with macro-F1, per-class F1, precision and recall, and summarize results in a single comparison table.

- **Modality ablation study**: Systematically remove each modality, measure the drop in macro-F1 to quantify the marginal contribution of each modality.

- **Early vs late fusion comparison**: Compare the two fusion strategies to determine which fusion strategy better leverages cross-modality information.

- **Interpretability**: Use SHAP values [3] for the best tree-based model and contrast with LASSO coefficients to identify the most discriminative features across tumor types.

- **Class-wise error analysis**: Use confusion matrices to identify the most confused class pairs and investigate which modalities/features differentiate them.

· **Imbalance strategy comparison**: Report the effect of each resampling strategy on the performance of minority classes.

## 5. Timeline

| Period | Key Tasks |
|---|---|
| Apr 1–5 | Proposal; data pipeline finalized; LR baseline; first Kaggle submission |
| Apr 6–8 | Single-modality baselines (Stage 1); 5-fold CV framework |
| Apr 9–11 | Multimodal fusion (Stage 2); feature selection and imbalance experiments (Stage 3) |
| Apr 12–13 | Best model selected; final Kaggle submission; experiment log frozen |
| Apr 14–20 | Report writing; SHAP & ablation analysis; clean Jupyter notebook |
| Apr 21–23 | Slides preparation; rehearsal |
| Apr 24 | Oral presentation |

## 6. Work Distribution

| Member | UID | Responsibility |
|---|---|---|
| Zhou Yuxuan | 3036291813 | Data pipeline, radiomics processing, traditional ML baselines (RF, SVM) |
| Cheng Hengqi | 3036292427 | Deep image features, PCA tuning, MLPs experiments |
| Ye Leyi | 3035973971 | Text/NLP feature extraction, text-model experiments |
| Shen Hongshan | 3036290936 | Multimodal fusion, ensemble methods, hyperparameter tuning |
| Qin Zilan | 3036290003 | CV framework, model analysis (SHAP, ablation), report and slides |

## References


[1] Price M, Ballard C, Benedetti J, et al. CBTRUS statistical report: primary brain and other central nervous system tumors diagnosed in the United States in 2017–2021[J]. *Neuro-oncology*, 2024, 26(Suppl 6): vi1.

[2] Wang Y R J, Wang P, Yan Z, et al. Advancing presurgical non-invasive molecular subgroup prediction in medulloblastoma using artificial intelligence and MRI signatures[J]. Cancer Cell, 2024, 42(7): 1239-1257. e7.

[3] He K, Zhang X, Ren S, et al. Deep residual learning for image recognition[C]//Proceedings of the IEEE conference on computer vision and pattern recognition. 2016: 770-778.

[4] Lundberg S, Lee S. A unified approach to interpreting model prediction[C]//Advances in Neural Information Processing Systems. 2017.